EXPERIMENT-5 Develop a customized ping command to test the server connectivity

Aim

To develop a customized Ping command in Python that checks the connectivity between a client and a server by sending ICMP Echo Request packets and displaying the response time and status.

Requirements

Hardware

- Computer/Laptop
- Internet Connection

Software

- Windows 10/11 or Ubuntu Linux
- Python 3.10 or above
- Visual Studio Code (VS Code) or PyCharm
- Command Prompt (Windows) / Terminal (Linux)

Theory

A Ping command is a network diagnostic tool used to determine whether a host (server) is reachable over a network.

It works using the ICMP (Internet Control Message Protocol).

The customized ping program allows the user to:

- Enter any hostname or IP address.
- Send multiple ping requests.
- Display response time.
- Show packet loss.
- Indicate whether the server is reachable.

Working Procedure

1. Install Python and VS Code.
2. Create a Python file named ping.py.
3. Import the required modules (subprocess and platform).
4. Accept the hostname or IP address from the user.
5. Detect the operating system.
6. Use ping -n 4 on Windows or ping -c 4 on Linux/macOS. 7.

Execute the ping command using the subprocess module. 8. Display the ping output, including reply time and packet statistics. 9. Check the return code to determine whether the server is reachable. 10. Display **Reachable** or **Unreachable** based on the result. **Step 1: Install Python**

Windows

1. Open browser.
2. Visit

https://www.python.org/downloads/

3. Download the latest Python.
4. Install it.
5. Important: Enable

✔ Add Python to PATH

6. Click Install Now.



Step 2: Install VS Code

Download from

https://code.visualstudio.com/
Install normally.

Open VS Code.

Install **Python Extension**.

Step 3: Create Project Folder

Example

```
CustomizedPing
```

Inside create

```
ping.py
```

Step 4: Write the Program

```
import subprocess

import platform


print("="*50)

print(" Customized Ping Utility")

print("="*50)


host = input("Enter Website or IP Address: ")


system = platform.system()


if system == "Windows":

 command = ["ping", "-n", "4", host]

else:

 command = ["ping", "-c", "4", host]


try:

 result = subprocess.run(command, capture_output=True, text=True)
 print("\nPing Results\n")

 print(result.stdout)


 if result.returncode == 0:

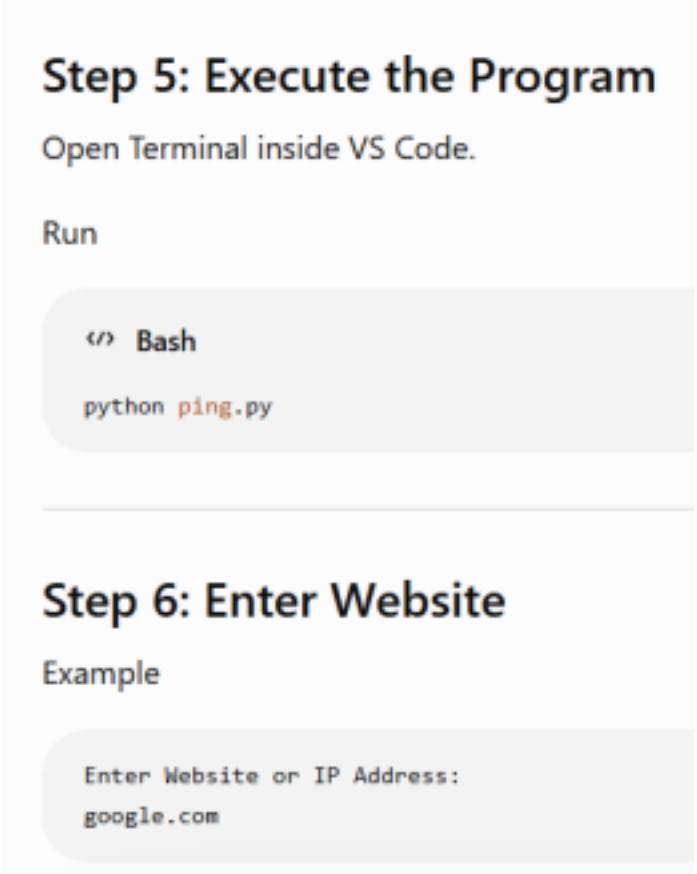
 print("Server Status : Reachable")

 else:
```

```
 print("Server Status : Unreachable")


except Exception as e:
 print("Error:", e)
```

Save the file.



Sample Output 1 (Successful)

```
================================================
          Customized Ping Utility
================================================

Enter Website or IP Address:
google.com

Pinging google.com [142.250.193.78] with 32 bytes

Reply from 142.250.193.78:
bytes=32 time=28ms TTL=116

Reply from 142.250.193.78:
bytes=32 time=30ms TTL=116

Reply from 142.250.193.78:
bytes=32 time=27ms TTL=116

Reply from 142.250.193.78:
bytes=32 time=29ms TTL=116


Ping statistics for 142.250.193.78

Packets:
Sent = 4
Received = 4
Lost = 0 (0% loss)

Approximate round trip times

Minimum = 27ms
Maximum = 30ms
Average = 28ms
```

Sample Output 2 (Server Not Reachable)

Input

```
abcxyz123.com
```

Output

```
Ping request could not find host abcxyz123.com.

Server Status : Unreachable
```